

Skills checks cover pre-requisite material needed for the third unit of the course, applications of derivatives. Answer the following questions neatly and completely. Calculators are not allowed.

1. Given $3(x^2 + 4)^{1/3} - 2x(x^2 + 4)^{-2/3}$, factor out $(x^2 + 4)^{-2/3}$ and write your final answer using only positive exponents.
2. Given $\sqrt{x+1} - 5(x+1)^{-1/2}$, factor out $(x+1)^{-1/2}$ and write your final answer using only positive exponents.
3. Given $7(2x-3)^{2/3} - x(2x-3)^{-1/3}$, factor out $(2x-3)^{-1/3}$ and write your final answer using only positive exponents.
4. Solve the inequality $\frac{2x+1}{x^2} \geq 0$. Give your answer in interval notation.

5. Solve the inequality $(x-2)(x+1)^3 > 0$. Give your answer in interval notation.

6. Solve the inequality $e^x(x-2) \leq 0$. Give your answer in interval notation.

7. Solve the equation: $x \ln(2) - x^2 = 0$

8. Solve the equation: $e^x - xe^x = 0$

9. Solve the equation: $x^2 \ln(x) - 4 \ln(x) = 0$.

10. Solve the equation $6 \tan(\theta) - 6 = 0$ for θ .

11. Solve the equation $\frac{1}{2} + \cos(\theta) = 0$ for θ .

12. Solve the equation $2 \sin(\theta) - 1 = 0$ for θ .

13. Find the volume of a cylinder with height of 5cm and a diameter of 2cm .

14. Find the radius of a circle that has an area of 12in^2 .

15. Determine the length of the remaining side of the triangle at the right.

